

Inventory Management & Inventory Accuracy Recovery

Rebuilding trust in inventory data across a two country field service spare parts network

87 → 98%

INVENTORY
ACCURACY

\$44.2K

INVENTORY
VALUE

826

UNITS TRACKED

3

VARIANCES
RESOLVED

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SECTION 1

Executive Summary

The problem

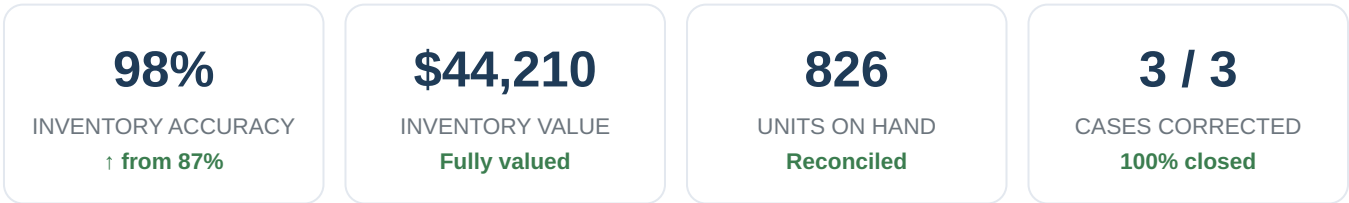
Inventory accuracy across a multi site ATM parts distribution network had deteriorated to around **87%**. Stock discrepancies, unrecorded consumption, damaged stock, and weak location control meant the business could no longer trust its own inventory data. The knock on effects were emergency purchases, duplicate replenishment, engineers arriving on site without parts, and slow manual reconciliation at month end.

What was done

I designed and configured a Dynamics 365 Supply Chain inventory management solution covering the full operating model: a multi site warehouse structure with bin level locations, product and costing controls, serial and batch tracking, and controlled inventory transactions. I then ran a structured inventory accuracy investigation, tracing each variance to its root cause and correcting it through the appropriate journal. Finally, I built a Power BI analytics layer to make accuracy measurable and visible.

The outcome

Inventory accuracy was restored from **87% to 98%**, supported by real time inventory visibility, structured transaction control, and executive reporting. Every one of the three investigation cases was traced to root cause and closed. The same data model now provides the foundation for a future inventory intelligence agent.



The headline insight

Inventory accuracy is not a counting problem. In this project, the variances traced back to transactions that were never posted, stock that was damaged in storage, and stock recorded against the wrong bin. The fix was transaction discipline and structured control, not more counting.

Environment note. Built in a Dynamics 365 Supply Chain demo environment on the standard USMF entity, with a dedicated two country site and warehouse network and a purpose built set of 30 spare parts. This keeps the focus on the inventory accuracy story rather than base company setup.

SECTION 2

Business Context

A field service organisation maintains critical customer equipment, primarily ATMs and banking hardware, across Nigeria and Ghana. Its ability to complete a repair depends on having the right spare part, in the right place, at the right time.

Spare parts are held across a tiered network of stocking points. A central warehouse in Lagos acts as the primary distribution hub. Regional warehouses in Abuja and Accra support their local territories. Field engineer stores sit close to the engineers who consume the parts. The catalogue spans high value serviceable modules, fast moving consumables, and serialized critical components.

Network element	Role
Lagos Central Warehouse (LAG-CW)	Primary distribution hub, holds the majority of network value
Abuja Regional Warehouse (ABJ-RW)	Serves the northern Nigeria territory
Accra Regional Warehouse (ACC-RW)	Serves the Ghana operation
Field Engineer Stores	Boot stock held close to the point of consumption

Warehouse Network

Two country field service spare parts distribution network

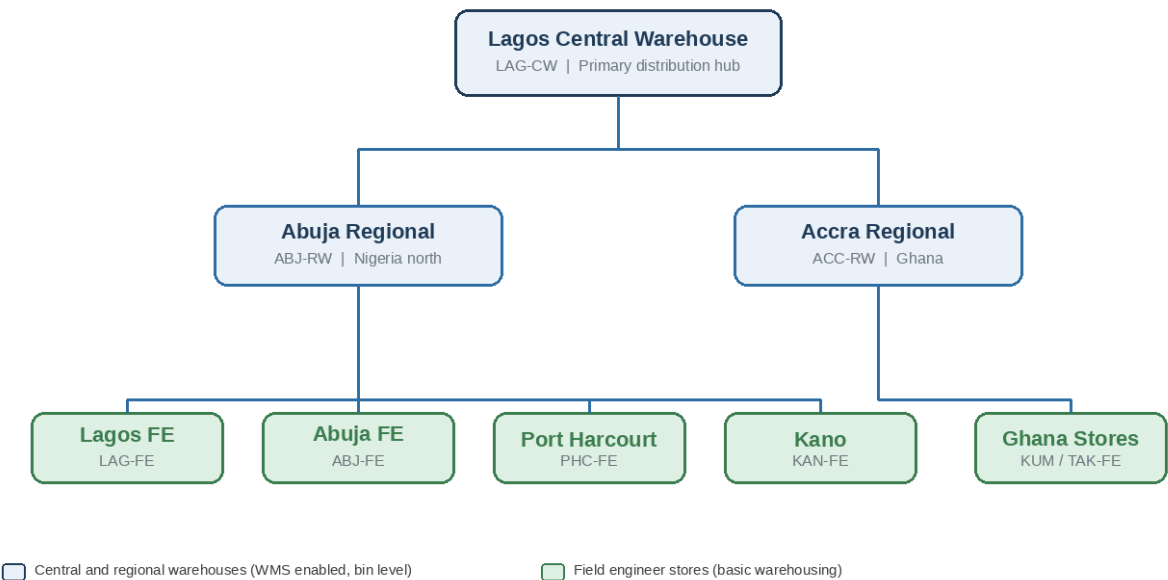


Figure 2.1 — Warehouse network. Central and regional warehouses run WMS enabled bin level control; field stores use basic warehousing.

An internal review found inventory accuracy had fallen to around **87%**. In practice this meant that for a meaningful share of items the system quantity did not match the physical shelf. Because planning, replenishment, and service delivery all depend on that data, the impact rippled across the whole operation.

Symptoms observed

- Stock discrepancies between the system and the physical shelf.
- Unrecorded consumption when parts were fitted on site but never issued in the system.
- Damaged stock that remained on the books until a physical inspection found it.
- Misplaced inventory sitting in the wrong bin, breaking location level accuracy.
- Emergency purchases and duplicate replenishment driven by unreliable data.

SECTION 3

Business Requirements

Objectives

- Restore inventory accuracy to a 98% target, measured against physical counts.
- Establish real time inventory visibility by site, warehouse, and bin.
- Route every stock movement through controlled, auditable journals.
- Reduce manual reconciliation effort at month end.
- Provide management reporting that makes accuracy and value measurable.

Scope

In scope	Out of scope
Product, costing, and tracking configuration	Finance and ledger design beyond posting defaults
Multi site warehouse and bin structure	Advanced WMS wave and load processing
Inventory transactions and journals	Full procure to pay receipting cycle
Inventory accuracy investigation and recovery	Demand forecasting and master planning
Power BI analytics layer	Production and manufacturing execution

Success criteria

- Inventory accuracy demonstrably recovered to the 98% target.
- Every variance traced to a documented root cause and corrected through the right journal.
- On hand balances reconciled across the network and visible to bin level.
- A working analytics dashboard reporting accuracy, value, and variance.

Assumptions

- A representative product set of 30 SKUs stands in for the full catalogue.
- Central and regional warehouses use WMS enabled locations; field stores use basic warehousing.
- Unit costs are treated in USD for valuation consistency.
- Accuracy is reported as a case study KPI, since a computed figure needs a defined counted population.

SECTION 4

My Role

I led this engagement end to end as the Dynamics 365 SCM functional consultant, from process design through configuration, testing, investigation, and analytics. The list below maps directly to the skills a hiring manager can expect me to bring.

What I delivered

- Designed the two country warehouse network structure and control model.
- Configured inventory parameters in Dynamics 365 SCM, products, groups, dimensions, and locations.
- Created and tested inventory transactions across movement, transfer, and procurement.
- Investigated inventory discrepancies to root cause and corrected them through the right journals.
- Built the Power BI reporting layer, from data model to four dashboard pages.
- Produced the process documentation and step by step SOPs behind the build.

Skills demonstrated

Area	Capability
Solution design	Process mapping, control point design
D365 SCM	Product, warehouse, and inventory configuration
Operations	Transaction execution and root cause investigation
Analytics	Power BI data modelling and dashboard design
Documentation	SOPs, process maps, evidence packs

In one line

I bridged the gap between the operation and the system: understanding the business problem, configuring Dynamics 365 to solve it, and building the analytics to prove it.

SECTION 5

Solution Architecture

The solution was designed as a process before it was configured as a system. Mapping the flows first kept the configuration purposeful and made the control points explicit.

End to end inventory flow

The operating model follows a single controlled path from supplier to reporting. Each stage writes a traceable record, and each control point is where accuracy is either protected or lost.

Stage	Control point
Vendor → Purchase Order → Receipt	Controlled inbound, receipt posted before put away
Warehouse Storage → Transfer	Bin level location control, traced site to site
Consumption → Counting	Issue recorded at point of use, variances surfaced by count
Adjustment → Reporting	Corrections through controlled journals, measured in Power BI

Core process maps

Three processes carry the inventory lifecycle. Receiving brings stock in under a quality gate. Consumption governs how stock leaves, the point where undocumented issues quietly erode accuracy. Adjustment is the control loop that protects accuracy over time through count, investigate, approve, and post.

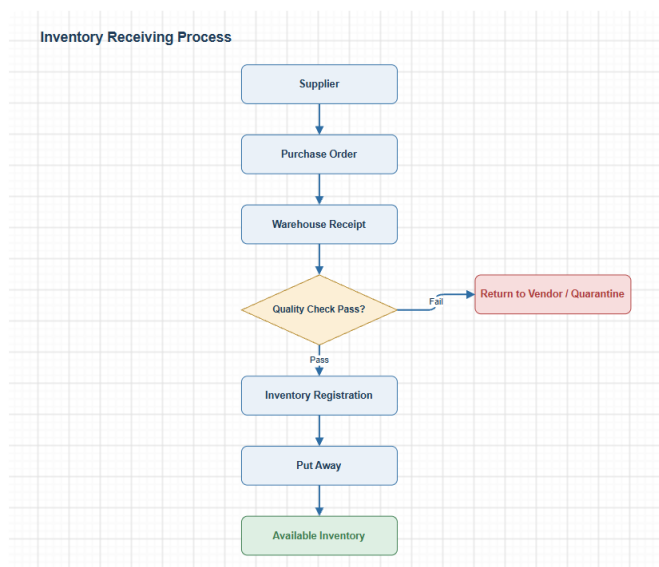


Figure 5.1 — Receiving process with quality gate.

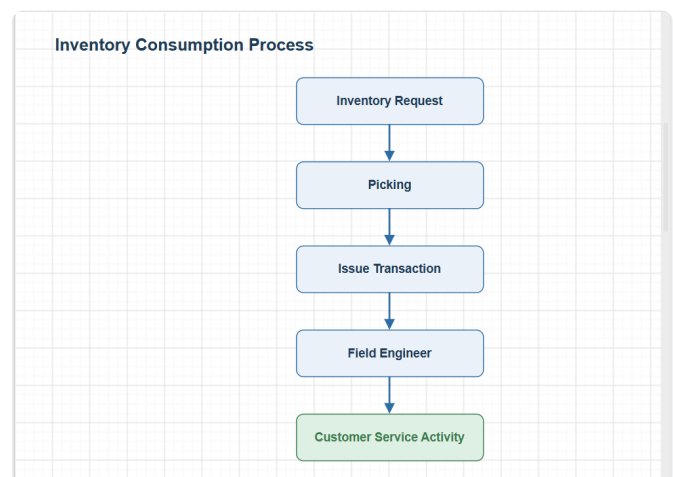


Figure 5.2 — Consumption process to point of use.

Inventory Adjustment Process

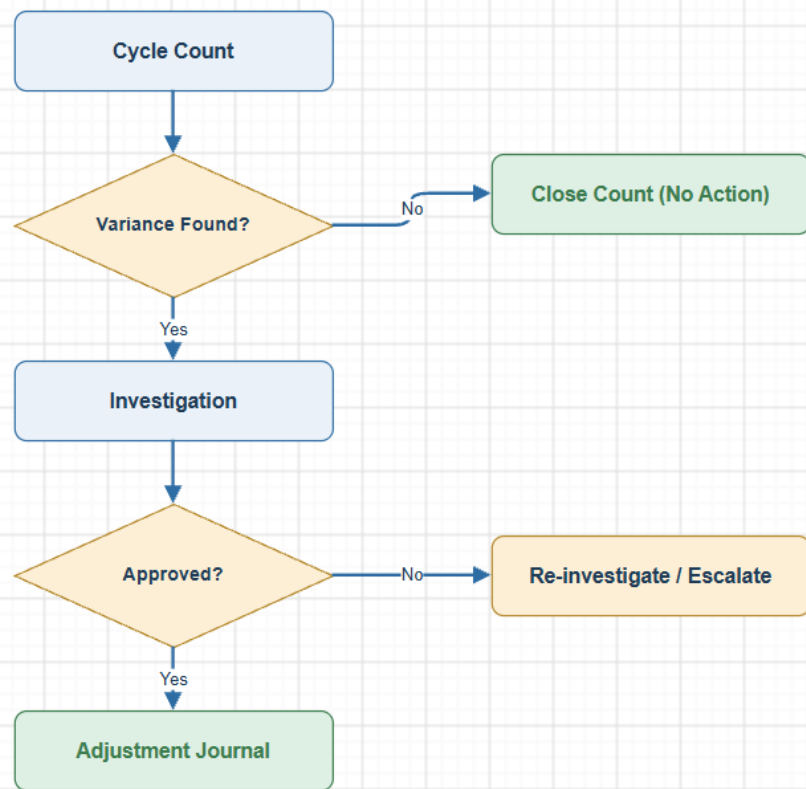


Figure 5.3 — Adjustment process, the count to correction control loop that sustains accuracy.

SECTION 6

Dynamics 365 SCM Configuration

Configuration is presented by purpose, not as a screen tour. The summary below gives the executive view; the evidence then follows area by area.

Configuration summary

Configuration area	Outcome
Product setup	30 inventory items configured across six categories
Warehousing	Multi site warehouse structure, two sites, eleven warehouses
Tracking	Serial and batch controls on the right item types
Locations	Bin level inventory tracking on the main warehouses
Procurement	Approved vendor framework aligned to the product master
Inventory control	Movement, transfer, and counting journals

6.1 Product structure

Purpose. Give every item a consistent identity for classification, costing, and traceability. Six product categories organise the catalogue, a single item group drives clean ledger posting, and serial tracking gives critical parts a per unit service history.

The screenshot shows the 'Category hierarchy' configuration page in Dynamics 365 SCM. The left sidebar lists the category hierarchy: 'New category' (selected), 'ATM Spare Parts', 'Display Modules', 'Network Components', 'Power Components', 'Core Computing', and 'Consumables'. The main area displays the configuration for the 'Consumables' category. The 'General' tab is active, showing fields for 'Name' (Consumables), 'Code' (CON), 'Active' (Yes), and 'Import status' (Not applicable). Below this, the 'General' section includes 'Created date' (8/29/2026 01:35:51 PM), 'Modified by' (Admin), 'Description' (Paper, ribbons, cleaning kits, seals), 'Friendly name' (Consumables), 'Keywords', and 'Category Display Order' (0.00). The 'Products' section is collapsed. The 'Category attributes' section shows 'Inherit category attributes from ...' set to 'No'. Below this is a table with columns: Name, Type, Attribute type, and Inherited from. The table is empty, with a message 'We didn't find anything to show here.' at the bottom. The 'Commodity codes' section is also collapsed.

Figure 6.1 — Product category hierarchy, six categories classifying all 30 items.



Purpose. Model the physical network so on hand is precise to the shelf. Two sites carry eleven warehouses, and the main warehouses hold a full aisle rack bin location structure.



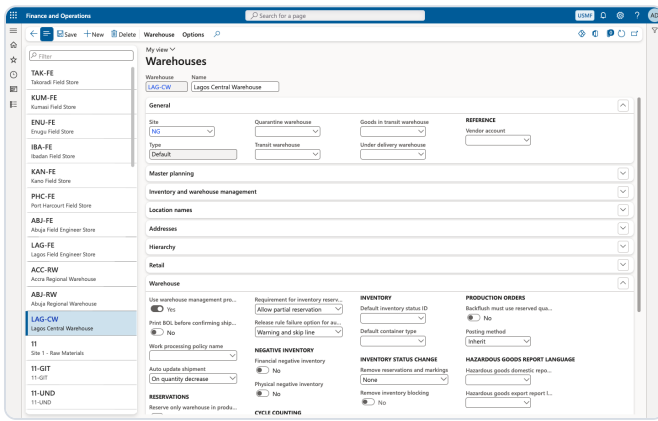


Figure 6.5 — Eleven warehouses across both sites.

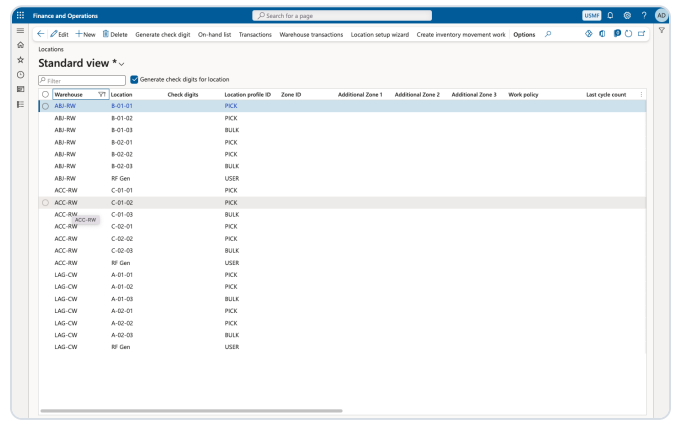


Figure 6.6 — Bin level locations, aisle rack bin.

6.3 Inventory controls

Purpose. Set the dimensions that make accuracy possible. Storage dimensions track stock to bin level, batch tracking governs consumables and batteries, and default order settings make transactions default to the correct site and warehouse.

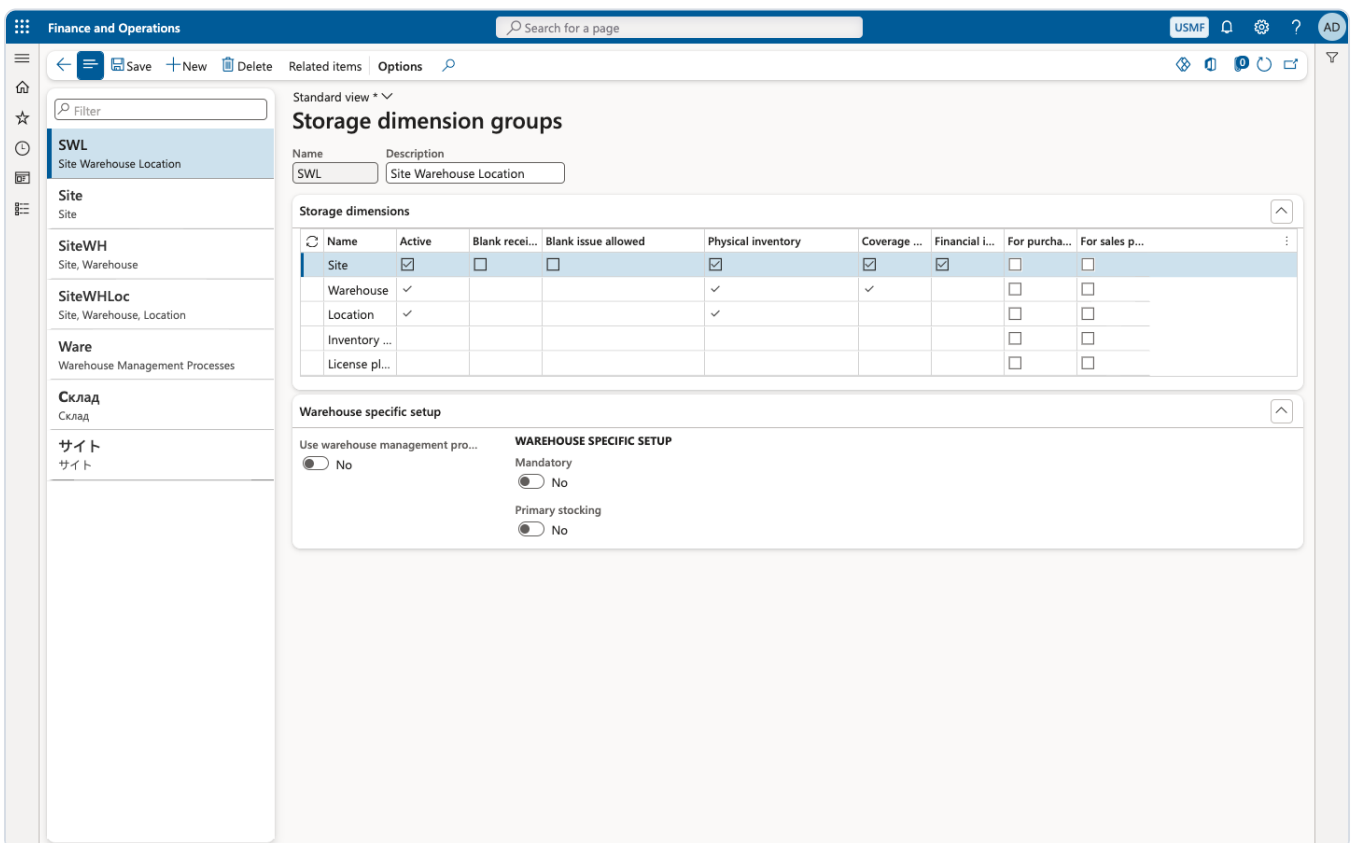


Figure 6.7 — Storage dimension group SWL, site, warehouse, location active.

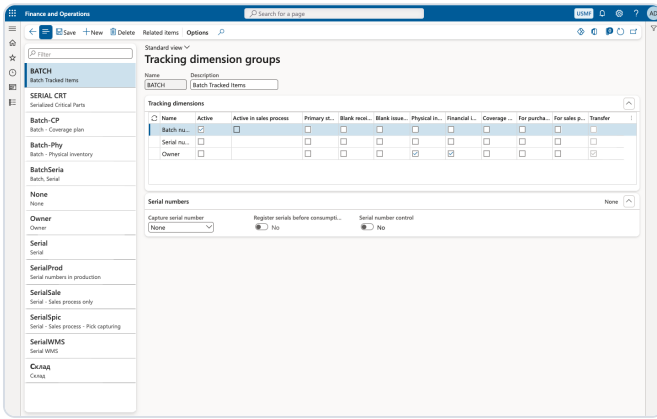


Figure 6.8 — Batch tracking group for consumables.

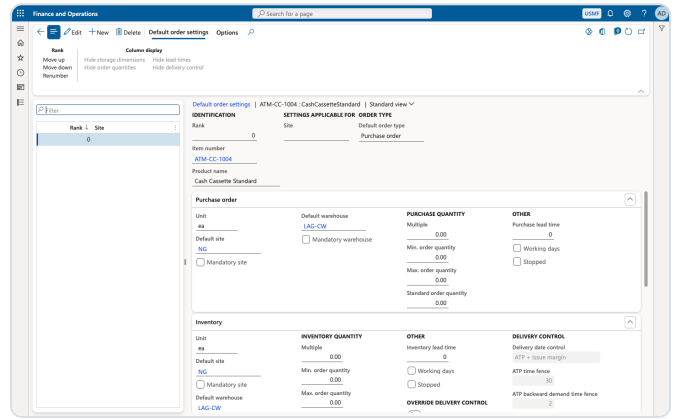


Figure 6.9 — Default order settings for clean defaults.

SECTION 7

Transaction Testing

With the model configured, I ran the transactions the operation depends on, capturing evidence at each step. Each transaction writes to the inventory tables that later feed analytics.

7.1 Opening inventory

Opening balances were loaded through a single controlled movement journal, giving a clean and valued starting position across the network. On hand was then validated against the seed.

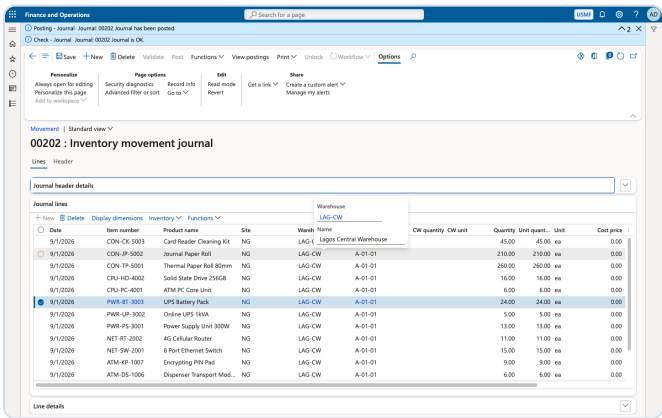


Figure 7.1 — Movement journal posted, opening stock seeded.

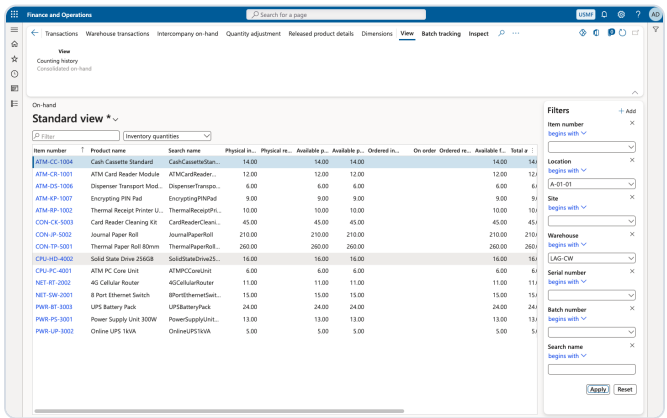


Figure 7.2 — On hand validation after seeding.

7.2 Warehouse replenishment

Stock was moved from Lagos Central to the regional warehouses through a controlled transfer, traced from origin bin to destination bin. On hand at the destination confirms the movement balances end to end.

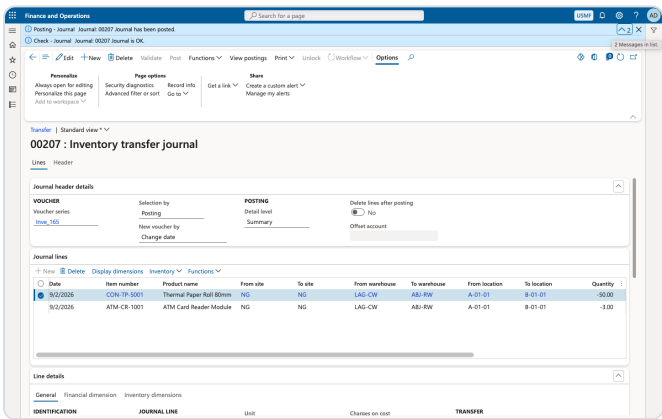


Figure 7.3 — Inventory transfer journal posted.

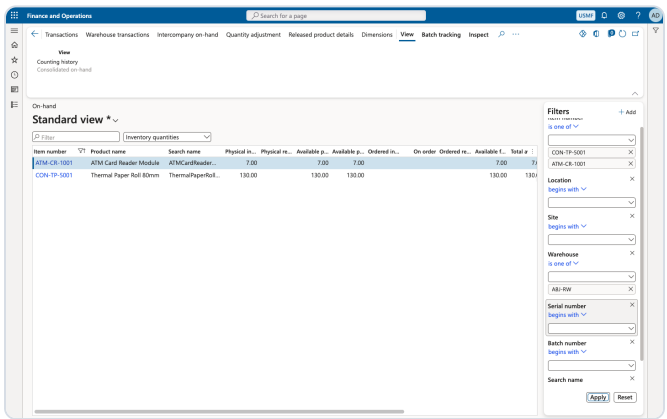


Figure 7.4 — On hand at destination after transfer.

7.3 Procurement

Replenishment from approved vendors was raised through a purchase order and formally confirmed. Confirming an order books it as ordered stock rather than physical stock, so on hand only rises on receipt, a useful control distinction.

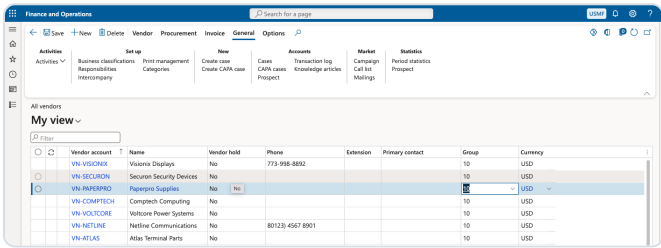


Figure 7.5 — Approved vendors, aligned to the product master.

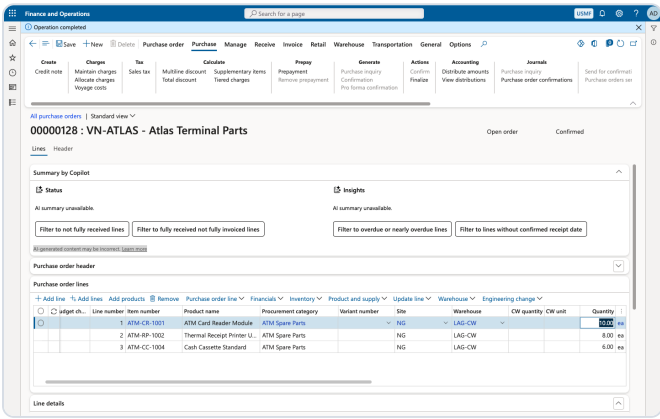


Figure 7.6 — Purchase order confirmed, ready for receipt.

SECTION 8

Inventory Accuracy Recovery

This is the flagship section. It is where the work moves from configuring a system to thinking like a consultant. Accuracy had fallen from a healthy baseline, and each variance was investigated to root cause, then corrected through the appropriate journal.

Investigation 1 — Unrecorded consumption

- Problem**

A physical count of ATM Card Reader Modules at Abuja showed fewer units than the system held.
- Root cause**

Parts were collected and fitted on site during an emergency repair, but the inventory issue was never recorded. A process failure, not a system failure.
- Action**

A counting journal captured the true quantity and posted the variance, correcting the balance.
- Result**

On hand realigned to the physical shelf; the control gap was identified as issue at point of use.

Investigation 2 — Damaged inventory

- Problem**

A monthly inspection at Lagos Central found damaged UPS batteries still counted as good stock.
- Root cause**

Storage failure. Units were physically unusable but remained on the books until inspection.
- Action**

An adjustment journal wrote off the damaged units with batch traceability, then posted.
- Result**

Inventory value and quantity corrected to reflect usable stock only.

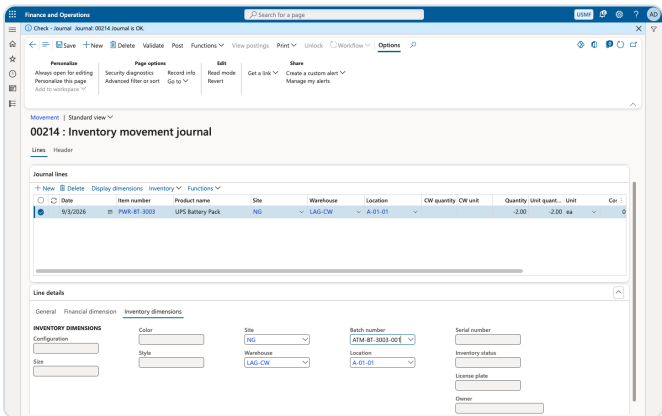


Figure 8.1 — Damaged stock adjustment validated, batch traceable.

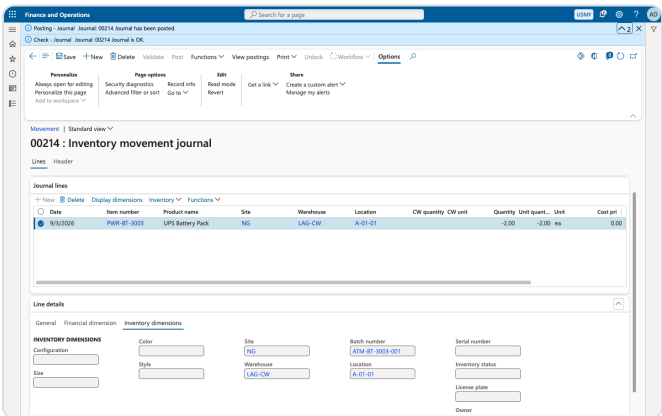


Figure 8.2 — Adjustment posted, balance corrected.

Investigation 3 — Misplaced inventory

Problem	During cycle counting, an Ethernet Switch was found physically in bin A-02-01 while the system showed it in A-01-01.
Root cause	Location control failure. The total was right but the bin was wrong, breaking location level accuracy.
Action	A transfer journal moved the unit to the correct bin, restoring location accuracy.
Result	Bin level on hand matched the physical shelf; the fix was tighter location discipline on counts.

Root cause analysis

Issue	Root cause	Corrective action
Unrecorded consumption	Process failure	Counting journal
Damaged inventory	Storage failure	Adjustment journal
Misplaced inventory	Location control failure	Transfer journal

Why this matters

Three different failure modes, three different corrective mechanisms. Recognising which control failed, and applying the matching journal, is what sustains accuracy. Counting harder would not have fixed any of these on its own.

SECTION 9

Power BI Analytics

The final layer turns the inventory data into insight. A clean star schema feeds four dashboard pages that make accuracy, value, variance, and activity measurable at a glance. Each page below states its purpose, the key insight, and the business decision it enables.

Live interactive dashboard

Explore the full report in the browser, no sign in required: [Open the interactive Power BI report](#).

Data sources and model

Source	Role in the model
Transactions	Every movement, classified by activity type and direction
Products	Product dimension with cost, category, and reorder data
Warehouses and Sites	Network dimensions for site and warehouse analysis
On Hand	Current stock position and extended value
Variance Log	The three accuracy investigation cases

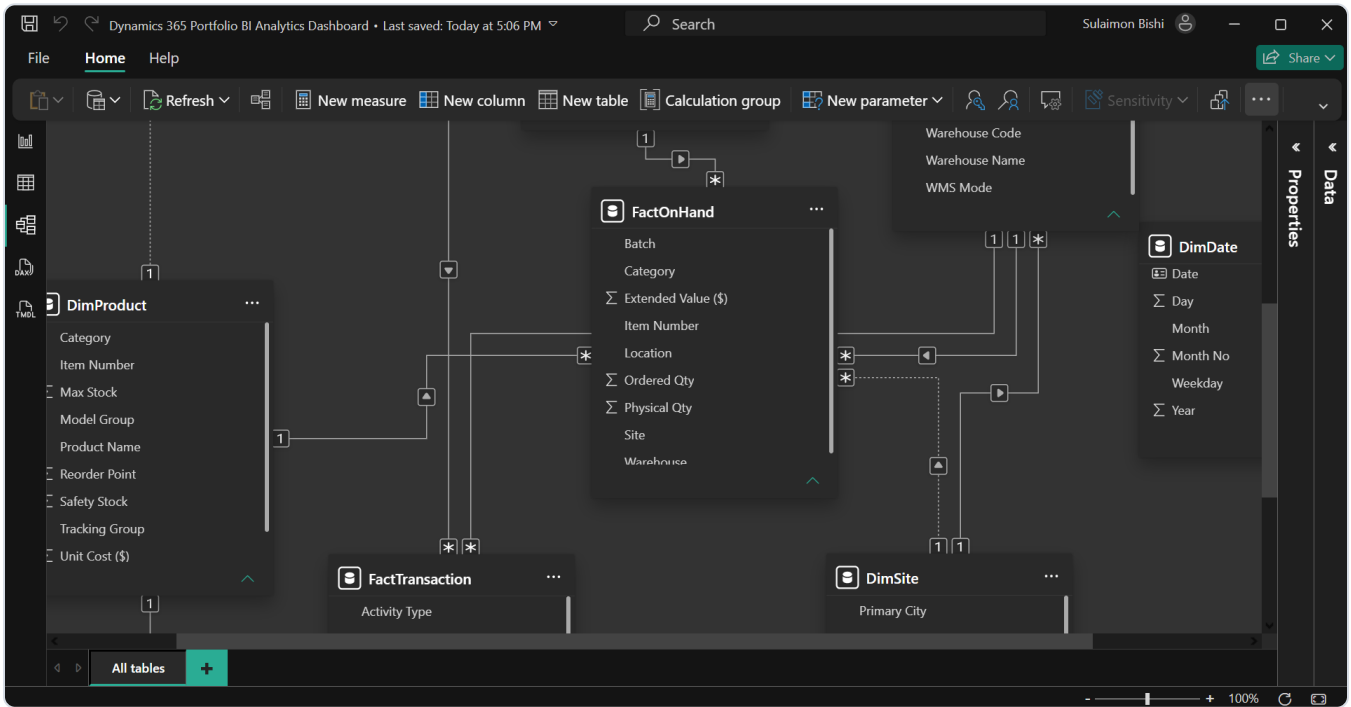


Figure 9.1 — Power BI data model, a star schema with dimensions on the one side and facts on the many side.

Dashboard 1 — Executive Overview

Purpose	Key insight	Decision enabled
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One glance health of inventory accuracy, value, and variance

Accuracy recovered to 98%,
\$44.21K value across 826 units

Confirms the recovery target is met and inventory data can be trusted again



Figure 9.2 — Executive Overview, headline KPIs and accuracy recovery.

Dashboard 2 — Warehouse Operations

Purpose	Key insight	Decision enabled
Show where inventory value concentrates across the network	Lagos Central holds the majority of value; ATM Spare Parts are half of it	Focus control effort and cycle counting on the highest value warehouse and category



Figure 9.3 — Warehouse Operations, value by warehouse, category, and item.

Dashboard 3 — Inventory Accuracy Recovery

Purpose	Key insight	Decision enabled
Make the investigation traceable and measurable	Unrecorded consumption is the largest variance source; all cases closed	Prioritise the issue at point of use control that prevents the biggest loss



Figure 9.4 — Accuracy Recovery, variance by cause, warehouse, and status.

Dashboard 4 — Transaction Activity

Purpose	Key insight	Decision enabled
Expose the operational engine behind the numbers	Receipts dominate volume; replenishment is concentrated on critical ATM parts	Align procurement and replenishment planning to the fastest moving critical items

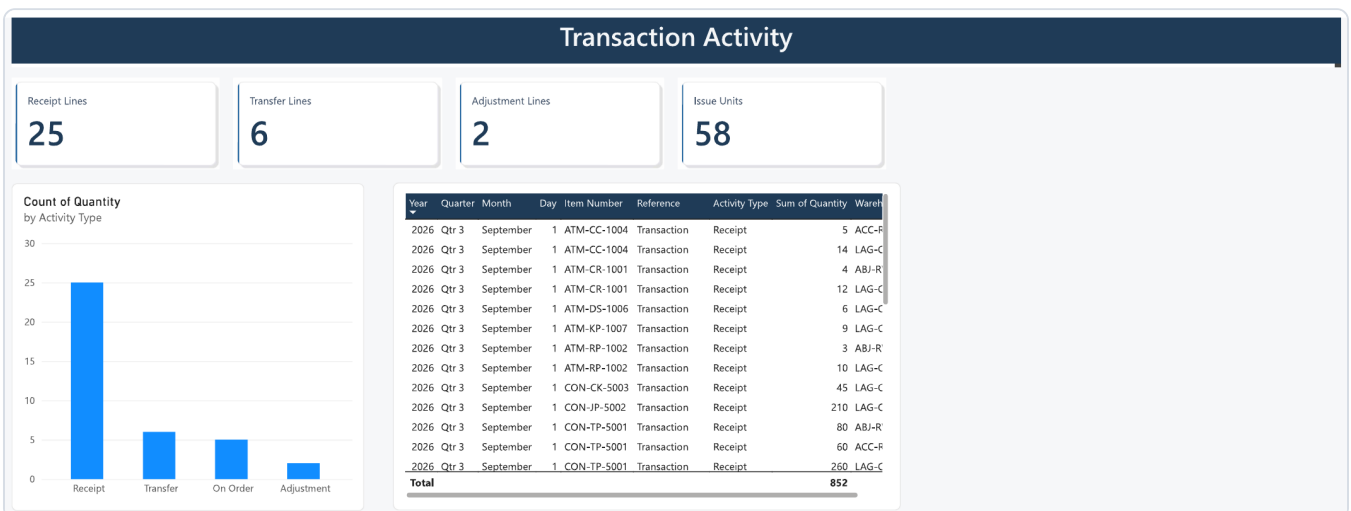


Figure 9.5 — Transaction Activity, movement counts and full audit trail.

SECTION 10

Business Outcomes

The measure of the project is business impact, not system features. The operation moved from unreliable, manual, spreadsheet driven inventory to a controlled, visible, and measured one.

KPI	Before	After
Inventory accuracy	87%	98%
Inventory visibility	Limited	Real time, to bin level
Inventory control	Manual	Structured, journal based
Reporting	Spreadsheet driven	Power BI analytics
Variance resolution	Undocumented	Root cause traced, 3 of 3 closed

Net result

Accuracy restored to the 98% target, inventory fully valued at \$44,210 across 826 units, and every variance closed. The business can once again trust its inventory data to plan, replenish, and serve customers.

Where the value shows up

- **Fewer emergency purchases**, because on hand data can be trusted.
- **Less duplicate replenishment**, because visibility is real time across the network.
- **Faster month end**, because corrections run through controlled journals rather than manual reconciliation.
- **Better service**, because engineers arrive with the parts the system promised.

SECTION 11

Consultant Recommendations

Recovery is the starting point, not the finish line. To hold accuracy at target and build on it, I would sequence the following, from quick wins to strategic capability.

Immediate — embed the discipline

- Make issue transactions mandatory for engineer consumption, recorded at the point of use.
- Introduce a monthly cycle counting program focused first on the highest value warehouse and category.

Medium term — remove manual error

- Deploy barcode scanning to eliminate keyed entry mistakes at receipt, transfer, and issue.
- Move warehouse transactions onto mobile devices so stock is updated as it physically moves.

Long term — move from reactive to predictive

- Introduce predictive replenishment that signals reorder before a stockout, using consumption trends.
- Add automated variance detection that flags divergence between system and physical stock continuously.

The path this sets

Each tier compounds the last. Discipline stabilises accuracy, automation protects it, and prediction turns inventory management from a corrective task into a strategic advantage.

SECTION 12

Lessons Learned

A short consultant reflection, the thinking that turns a build into a capability.

Key lessons

- **Inventory accuracy is primarily a process problem.** The system exposes the problem; the process is where it is created and solved.
- **ERP enforces control but does not replace discipline.** Dynamics 365 makes the right path easy, but people still have to post at the point of activity.
- **Analytics provides visibility into operational performance.** A dashboard turns scattered transactions into a management view of accuracy and value.
- **Root cause analysis is critical for sustainable control.** Matching each failure mode to the right corrective journal is what keeps accuracy from drifting again.

The consultant takeaway

The most valuable output was not the configuration, it was the diagnosis. Understanding why accuracy failed, and designing the control to prevent recurrence, is the difference between operating a system and improving an operation.

SECTION 13

Future State — AI Opportunity

The data this project generates makes an intelligent agent possible. This is the bridge from ERP delivery to operations intelligence.

Inventory Intelligence Agent

Today, when accuracy drops, an analyst spends days reconciling transactions, journals, and counts to find the cause. An inventory intelligence agent would do that reasoning in minutes, reading the same data model built here.

Capability	What it does
Automated variance detection	Continuously flags where system and physical stock diverge
Root cause recommendations	Classifies each variance by likely cause and suggests the corrective journal
Replenishment forecasting	Signals where stock will fall below reorder point before it stocks out
Warehouse exception monitoring	Surfaces abnormal warehouses and high variance SKUs for review

Connecting to the bigger vision

This agent concept is the first building block of an Operations Intelligence capability: using ERP data, analytics, and applied AI to move supply chain teams from reacting to problems to anticipating them.

APPENDIX

Evidence Index

This case study is a curated narrative, backed by a full evidence pack of annotated screenshots captured directly from the Dynamics 365 environment and the Power BI report. The index below maps each evidence ID to its content.

Evidence ID	Description	Section
CFG-01	Product categories	6.1
CFG-02	Item group SVC-PARTS	6.1
CFG-03	Serial and batch tracking groups	6.1 / 6.3
CFG-04	Sites NG and GH	6.2
CFG-05	Warehouses, eleven stocking points	6.2
CFG-06	Bin level warehouse locations	6.2
CFG-07	Storage dimension group SWL	6.3
CFG-08	Default order settings	6.3
TRX-01	Movement journal, opening stock seed	7.1
TRX-02	On hand validation after seeding	7.1
TRX-03	Transfer journal, inter warehouse	7.2
TRX-04	Approved vendors and purchase order	7.3
ACC-01	Count journal variance, unrecorded consumption	8
ACC-02	Inventory adjustment, damaged stock	8
ACC-03	Transfer correction, misplaced inventory	8
BI-01	Power BI data model	9
BI-02 to 05	Executive, Warehouse, Accuracy, Transaction dashboards	9

Environment and data

- **Platform.** Microsoft Dynamics 365 Supply Chain Management, USMF demo entity.
- **Network.** Two sites, eleven warehouses, bin level locations on the main warehouses.
- **Catalogue.** Thirty representative spare parts across six categories.
- **Analytics.** Power BI star schema across on hand, transactions, and a variance log.
- **Live report.** Published to the Power BI Service, public view. [Interactive dashboard link](#).

About the author

Sulaimon Oladipo Bishi is a supply chain, logistics, and operations professional and a Dynamics 365 SCM functional consultant, bridging operations and system configuration across Nigeria and Ghana. This case study is part of a wider portfolio building towards an Operations Intelligence practice.